DESCRIPTION OF COURSEWORK

Course Code	EEE411
Course Name	Communication System
Lecturer	Dr. Khuzairi Bin Masrakin
Academic Session	2024/09
Assessment Title	Time- and Frequency-domain Analysis

A. Introduction/ Situation/ Background Information

A time-domain graph shows how the amplitude of signal changes over time. On the other hand, a frequency-domain graph shows how much of the signal lies within each given frequency band over a range of frequencies. Communication signals and systems can be characterized in both the time and frequency domains.

In mathematics, the discrete Fourier transform (DFT) converts a finite sequence of equally-spaced samples of a function into a same-length sequence of equally-spaced samples of the discrete-time Fourier transform (DTFT), which is a complex-valued function of frequency. The DFT is the most important discrete transform, used to perform Fourier analysis in many practical applications. DFT can be implemented in computers by numerical algorithms or even dedicated hardware. These implementations usually employ efficient Fast Fourier Transform (FFT) algorithms which is an implementation of the DFT that produces almost the same results as the DFT, but it is incredibly more efficient and much faster which often reduces the computation time significantly.

GNU Octave is a high-level language, primarily intended for numerical computations. It provides a convenient command line interface for solving linear and nonlinear problems numerically, and for performing other numerical experiments using a language that is mostly compatible with Matlab. GNU Octave is also freely redistributable software. GNU Octave may be redistributed and/or modified under the terms of the GNU General Public License (GPL) as published by the Free Software Foundation. The latest version of GNU Octave is available for download at https://www.gnu.org/software/octave/download.

GNU Octave has built-in signal processing functions which include the Fast Fourier Transform (FFT) algorithm that could be use to plot the spectrum of a signal. The detailed descriptions of all signal processing functions in GNU Octave are available at https://docs.octave.org/v8.1.0/Signal-Processing.html.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students can:

CLO 2 Analyse major communication subsystem

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:

i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

1. This is a group work (2 to 4 students each group) assignment.
2. There are **TWO** tasks in this assignment. Attempt **ALL**.
3. You are required to use GNU Octave / Matlab to complete this assignment.
4. Use the cover page provided.
5. Please attach the complete GNU Octave / Matlab scripts in the Appendix of your report.
6. Deadline: **01 December 2024, 11:59:59 pm**
7. The group leader should submit the PDF softcopy of your report to my email on khuzairi.masrakin@xmu.edu.my.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	GNU Octave / Matlab scripts	20
2.	Project report	80
	TOTAL	**100**

F. Task(s)

Task 1: Plotting Spectrum using FFT

Write a GNU Octave / Matlab script to plot the spectrum of a time domain signal. You may use the built-in `fft` function in GNU Octave / Matlab®. Your script should plot the magnitude scale in the unit of decibel-milliwatt (dBm). Discuss how your script works.

[*Hint:*

1. *You may use a known sinusoidal tone to verify your code. E.g., the power level of a 1 V peak voltage sinusoidal waveform is 10 dBm.*
2. *Zero padding the data before computing the FFT often helps to improve the accuracy of amplitude estimates.*]

Task 2: Time and Frequency Domain Waveform Analysis

A Matlab® MAT file containing **FIVE** time domain waveforms is given in Moodle learning platform for this course. The waveforms and their sampling specifications can be loaded into the GNU Octave / Matlab® environment using the command

```
load('filename.mat')
```

Below is the description of the variables.

`xt1 – xt5`: Waveform data

`fs`: Sampling rate of the waveform data

`n`: Number of sample points of each waveform data

`t`: Time variables

Carry out time and frequency domain analyses on these waveforms to estimate the type of these waveforms. Use your scripts created in Task 1 to analyze the waveform in the frequency domain.

Suggestion for Report Format:

1. Introduction
2. Objective(s)
3. Discussion and results of work done for the given tasks
4. Discussion on learning outcomes of this task
5. Conclusion

Warning: This assignment is an open-ended problem with many correct answers and ways to find the correct answer(s). Similar code/script will be treated as plagiarism.

APPENDIX 1

MARKING RUBRICS

Component Title	GNU Octave / Matlab scripts				**Percentage (%)**		20	
Criteria	**Score and Descriptors**						**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Fair (2)**	**Poor (1)**	**Very Bad (0)**		
Scripts for Task 1	Complete scripts without needing correction	Able to plot the spectrum but require minor correction	Able to plot the spectrum, but the scaling was not proper	Able to plot the spectrum, but no scaling	Relevant to the topic but not able to plot the spectrum	Not relevant to the topic	20	
						TOTAL	**20**	

Component Title	Group report				**Percentage (%)**		80	
Criteria	**Score and Descriptors**						**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Fair (2)**	**Poor (1)**	**Very Bad (0)**		
Introduction	The introduction stated the main topic and engages the reader	The introduction stated the main topic	The introduction stated the main topic but requires minor correction	The introduction stated the main topic but requires major correction	Short introduction with inadequate information	Not found in the report	5	
Objective	The objectives are clear and well-defined	The objectives are clear	Somewhat well-defined objective(s)	Poorly defined objectives	The objectives are not relevant	Not found in the report	5	
Results for Task 1	Well-presented results	Satisfactory results	Satisfactory results but requires minor improvement	Inadequate results	The majority of the results are missing	Not found in the report	5	
Results for Task 2 (Waveform #1)	Well-presented results	Satisfactory results	Satisfactory results but requires minor improvement	Inadequate results	The majority of the results are missing	Not found in the report	5	
Results for Task 2 (Waveform #2)	Well-presented results	Satisfactory results	Satisfactory results but requires minor improvement	Inadequate results	The majority of the results are missing	Not found in the report	5	
Results for Task 2 (Waveform #3)	Well-presented results	Satisfactory results	Satisfactory results but requires minor improvement	Inadequate results	The majority of the results are missing	Not found in the report	5	
Results for Task 2 (Waveform #4)	Well-presented results	Satisfactory results	Satisfactory results but requires minor improvement	Inadequate results	The majority of the results are missing	Not found in the report	5	

Results for Task 2 (Waveform #5)	Well-presented results	Satisfactory results	Satisfactory results but requires minor improvement	Inadequate results	The majority of the results are missing	Not found in the report	5	
Discussion for Task 1	Excellent discussion	Good discussion	Satisfactory discussion	Fair discussion	Poor discussion	Not found in the report	5	
Discussion for Task 2 (Waveform #1)	Excellent discussion	Good discussion	Satisfactory discussion	Fair discussion	Poor discussion	Not found in the report	5	
Discussion for Task 2 (Waveform #2)	Excellent discussion	Good discussion	Satisfactory discussion	Fair discussion	Poor discussion	Not found in the report	5	
Discussion for Task 2 (Waveform #3)	Excellent discussion	Good discussion	Satisfactory discussion	Fair discussion	Poor discussion	Not found in the report	5	
Discussion for Task 2 (Waveform #4)	Excellent discussion	Good discussion	Satisfactory discussion	Fair discussion	Poor discussion	Not found in the report	5	
Discussion for Task 2 (Waveform #5)	Excellent discussion	Good discussion	Satisfactory discussion	Fair discussion	Poor discussion	Not found in the report	5	
Learning outcome	Well written	[redacted]	Satisfactory	[redacted]	Irrelevant	Not found in the report	5	
Formatting	Include all sections, neat and tidy report	[redacted]	Some section(s) was/were missing	[redacted]	More than half of the sections were missing	Not found in the report	5	
						TOTAL	**80**	

APPENDIX 2

Description of Complex Problem Solving (CPS)

The assignment is a complex engineering problem. Its mapping to the CLO and PLO are as follows.

1) Assignment 1 → CLO2 → PLO2

The complex problem-solving attributes involved in the final project are as follows.

Attribute	Remarks
WP1 Depth of knowledge required	The assignment is a complex engineering problem that cannot be resolved without in-depth knowledge of Fourier Analysis. The students need to possess in-depth knowledge of Fourier Analysis to develop a GNU Octave/Matlab script to perform Fast Fourier Transform (FFT) and use the script for time- and frequency-domain analysis.
WP2 Range of conflicting requirements	The assignment is to develop a script for frequency-domain analysis. The script involves a range of conflicting requirements, such as configuring the FFT, magnitude scaling, etc.
WP3 Depth of analysis required	The assignment is a problem-based design scenario. There is no obvious solution, and it requires abstract thinking. Each group has to propose a suitable hardware design with many conflicting requirements.